

AssignmentNumber:14.4

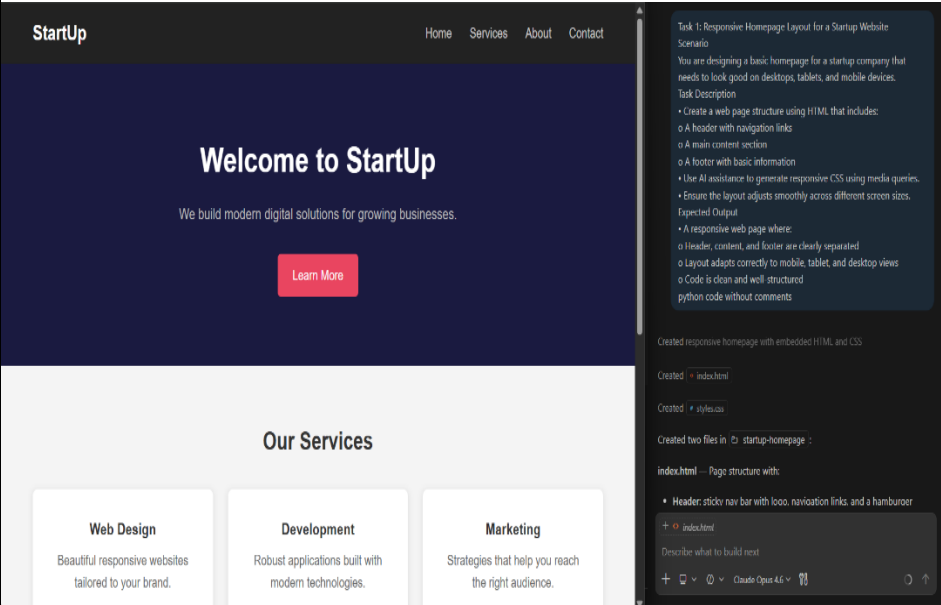
J.Telesh Hall-2303A51812 Batch-26

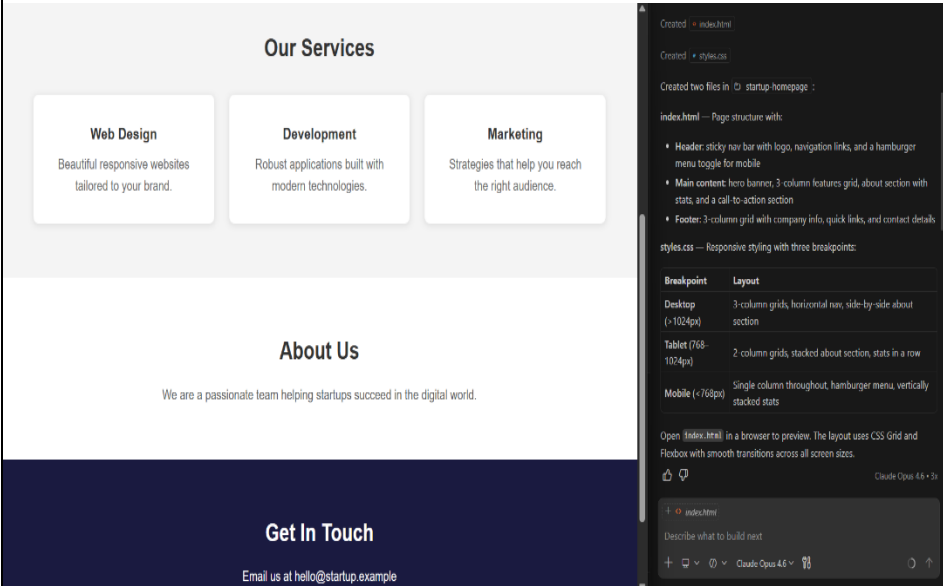
Q.No.	Question	ExpectedTime to complete
1	Lab 14: Web Frontend Development – AI-Assisted HTML/CSS/JS Generation Lab Objectives • Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance. • Apply responsive, accessible, and interactive design principles. • Create practical UI components for real-world web applications. • Use AI to optimize layout, UX, and performance. Lab Outcomes • Applying AI assistance to frontend development tasks • Understanding how HTML, CSS, and JavaScript work together • Improving responsiveness, interactivity, and usability • Evaluating and refining AI-generated frontend code	Week 7
	Task 1: Responsive Homepage Layout for a Startup Website Scenario You are designing a basic homepage for a startup company that needs to look good on desktops, tablets, and mobile devices. Task Description • Create a web page structure using HTML that includes: ◦ A header with navigation links ◦ A main content section ◦ A footer with basic information • Use AI assistance to generate responsive CSS using media queries.	

- Ensure the layout adjusts smoothly across different screen sizes.

Expected Output

- A responsive web page where:
 - Header, content, and footer are clearly separated
 - Layout adapts correctly to mobile, tablet, and desktop views
 - Code is clean and well-structured





Task 2: Interactive Call-to-Action Button

Scenario

A marketing page requires a **call-to-action button** that responds when users interact with it.

Task Description

- Add a button to a web page using HTML.
- Use AI to generate JavaScript that responds to the button click by:
 - Displaying a message or alert
- Ask AI to add meaningful comments explaining the JavaScript logic.

Expected Output

- A web page where:
 - Clicking the button triggers a visible response (alert or message)
 - JavaScript code is readable and commented
 - No page reload occurs

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	Task 3: Contact Form with Client-Side Validation Scenario You are developing a contact form for a company website to collect user queries safely and correctly. Task Description - Design a form with fields for: - Name - Email - Message - Use AI to generate JavaScript validation logic that: - Prevents empty submissions - Checks for a valid email format - Display clear error messages for invalid input. Expected Output - A functional form where: - Invalid input shows inline error messages - Valid input displays a success message	

- Validation happens on the client side using JavaScript

Get In Touch

Have a question? Send us a message and we'll get back to you.

Name

A.Shashidhar

Email

2303A51798@sru.edu.in

Message

Hello There!!!

Send Message

	Have a question? Send us a message and we'll get back to you. Name Your name Email you@example.com Message Your message Send Message Thank you! Your message has been sent successfully.	
	Task 4: Dynamic List Management for a Product Page Scenario A product listing page needs to add or remove items dynamically without refreshing the page. Task Description • Create a simple HTML list to display items. • Use AI to generate JavaScript that: ◦ Adds a new item when an “Add” button is clicked ◦ Removes the last item when a “Remove” button is clicked • Ensure the DOM updates dynamically.	

	Expected Output • A web page where: ○ Items can be added and removed dynamically ○ Changes appear instantly without page reload ○ JavaScript logic is clear and maintainable	
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Product List

AddRemove Last

Website Builder

SEO Toolkit

Analytics Dashboard

Product List

AddRemove Last

Website Builder

SEO Toolkit

Analytics Dashboard

laptop

	Product List Enter product name Add Remove Last Website Builder SEO Toolkit Analytics Dashboard	
	Task 5: Modal Popup for User Notifications Scenario You are building a user notification popup for announcements or alerts on a website. Task Description • Use AI to help generate: ◦ HTML structure for a modal popup ◦ CSS styling with a semi-transparent background overlay ◦ JavaScript to open and close the modal • Include multiple ways to close the modal (button or overlay click). Expected Output • A web page where: ◦ Clicking “Open Modal” displays the popup ◦ Popup overlays the page content clearly	

- Modal closes smoothly when instructed

Product List

AddRemove Last

Website Builder

SEO Toolkit

Analytics Dashboard

Open Modal

Product List

AddRemove Last

Website Builder

Notification

This is an important announcement! Check out our latest updates and new features now available on the platform.

Got it

Note: Report should be submitted a word document for all tasks in a

	single document with prompts, comments & code explanation, and output and if required, screenshots	
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